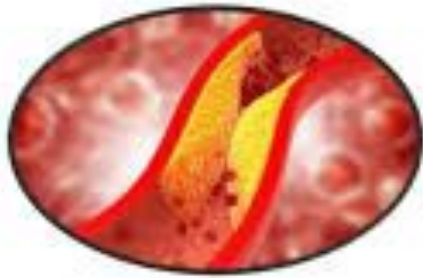


Cholesterol



CHOLESTEROL AND LIPID PEROXIDATION: MECHANISMS AND CLINICAL IMPLICATIONS

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INTRODUCTION TO LIPID PEROXIDATION

- Definition:**

The oxidative degradation of lipids. It is a process where free radicals "steal" electrons from lipids in cell membranes or lipoproteins.

- Target:**

Primarily Polyunsaturated Fatty Acids (PUFAs) and Cholesterol.

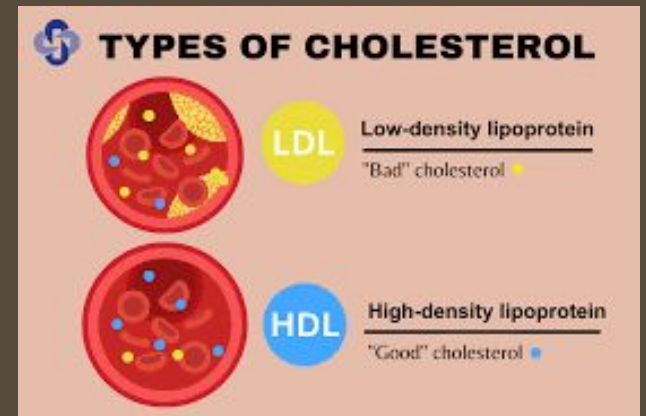
- Medical Significance:**

A key driver of aging, atherosclerosis, and membrane damage

When this happens, your cell membranes—the "security guards" of your cells—become leaky and brittle.

This leads to:

- Atherosclerosis: Damaged fats get stuck in artery walls, forming plaque.
- Aging: Accumulated "wear and tear" on cells.
- Disease: Linked to inflammation, cancer, and neurodegenerative disorders.
- The Bottom Line: Lipid peroxidation is the chemical "rusting" of your body's fats, leading to broken cell walls and internal inflammation.



MECHANISM – STEP 1: INITIATION

- **Trigger:** Reactive Oxygen Species (ROS) such as the hydroxyl radical ($\dot{\text{O}}\text{H}$).
- **Action:** The radical abstracts a hydrogen atom from a methylene group of a fatty acid or the cholesterol ring.
- **Result:** Formation of a **Lipid Radical (L)**.
- **Note:** This occurs most easily in the presence of transition metals like iron (Fe^{2+}).

MECHANISM – STEP 2: PROPAGATION

- **The Chain Reaction:** The lipid radical reacts with oxygen to form a **Lipid Peroxyl Radical (LOO[•])**.
- **Self-Sustenance:** This peroxy radical attacks an adjacent lipid molecule, creating a new lipid radical and a **Lipid Hydroperoxide (LOOH)**.
- **The Domino Effect:** This peroxy radical is highly unstable. To fix itself, it steals a hydrogen from a neighboring healthy lipid
- **Impact:** A single initiation event can damage hundreds of lipid molecules in a "domino effect". The first lipid becomes a Lipid Hydroperoxide (LOOH), but it has now created a new lipid radical in its neighbor. This cycle repeats thousands of times, "zipping" through the cell membrane and destroying its structure.

MECHANISM — STEP 3: TERMINATION

- Process:** The chain reaction ends when two radicals react to form stable, non-radical products.
- Natural Interruption:** Antioxidants like **Vitamin E** (Tocopherol) donate a hydrogen to the peroxy radical, neutralizing it before it can attack the next lipid.
- Biomarkers:** Final breakdown products include **Malondialdehyde (MDA)**(A byproduct (waste) used to measure how much damage has occurred), which is used clinically to measure oxidative stress.

THE PEROXIDATION FLOWCHART

Phase	Chemical Event	Key Molecular Actor
Initiation	Hydrogen abstraction	Hydroxyl Radical (.OH)
Propagation	Oxygen uptake & Chain reaction	Peroxyl Radical (LOO [•])
Degradation	Fragmentation of fatty acids	Malondialdehyde (MDA)
Termination	Radical scavenging	Vitamin E

CHOLESTEROL-SPECIFIC OXIDATION

- **Oxysterols:** When cholesterol is oxidized, it forms cholesterol oxidation products (COPs) or oxysterols (Oxysterols are Cholesterol Oxidation Products (COPs). They form when cholesterol molecules are attacked by free radicals or specific enzymes).
- Key Example: 7-ketocholesterol is one of the most common and toxic oxysterols found in human atherosclerotic plaques.
- Pure Cholesterol: A stable, necessary "brick" for your cells.
- Oxysterol: A "damaged brick" that is reactive and unstable.
- **Cytotoxicity:** These are significantly more toxic than pure cholesterol. Oxysterols are cytotoxic, they are directly poisonous to cells. While normal cholesterol stays quietly in the cell membrane, oxysterols can trigger cell death (apoptosis) and damage the cell's internal machinery.
- **Pathology:** Oxysterols inhibit the enzyme *HMG-CoA Reductase* (feedback inhibition) and are highly pro-inflammatory.



FUNCTION OF CHOLESTEROL:

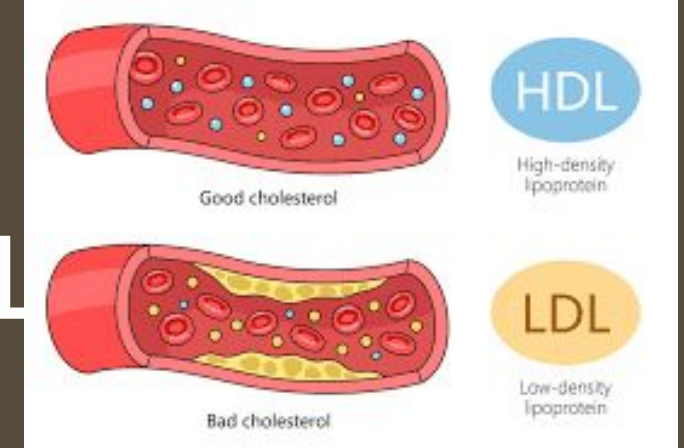
Cholesterol isn't used for energy like other fats; instead, it serves three vital roles:

Cell Membrane Structural Integrity: It acts like a "glue" or stabilizer for your cell walls, making them firm but flexible.

Hormone Production: It is the chemical starting point (precursor) for making Vitamin D and essential hormones like **estrogen, testosterone, and cortisol**.

Digestion: The liver uses cholesterol to create **bile acids**, which help you digest and absorb fats from your food.

NORMAL LDL VS. OXIDIZED LDL



Feature	Native LDL (Normal)	Oxidized LDL (Modified)
Receptor	ApoB-100 Receptor	Scavenger Receptor (SR-A)
Regulation	Feedback inhibited (SREBP)	No Feedback Inhibition
Cellular Fate	Used for membrane/hormones	Becomes a Foam Cell
Inflammation	Neutral	Highly Pro-inflammatory
Location	Systemic Circulation	Sub-endothelial space (Artery wall)

LIPID PROFILE

	DESIRABLE	BORDERLINE	HIGH RISK
Cholesterol	<200 mg/dl	200-239 mg/dl	240 mg/dl
Triglycerides	<150 mg/dl	150-199 mg/dl	200-499 mg/dl
HDL cholesterol	60 mg/dl	35-45 mg/dl	<35 mg/dl
LDL cholesterol	60-130 mg/dl	130-159 mg/dl	160-189 mg/dl
Cholesterol/ HDL ratio	4.0	5.0	6.0

THE ANTIOXIDANT "DEFENSE TEAM"

The Three Lines of Defense

1.Preventive (Enzymes): Superoxide Dismutase (SOD): Converts $\text{O}_2^\bullet$ (free radical) to H_2O_2 .

- Catalase:** Breaks down H_2O_2 into harmless water and oxygen.

2.Chain-Breaking (Vitamins):

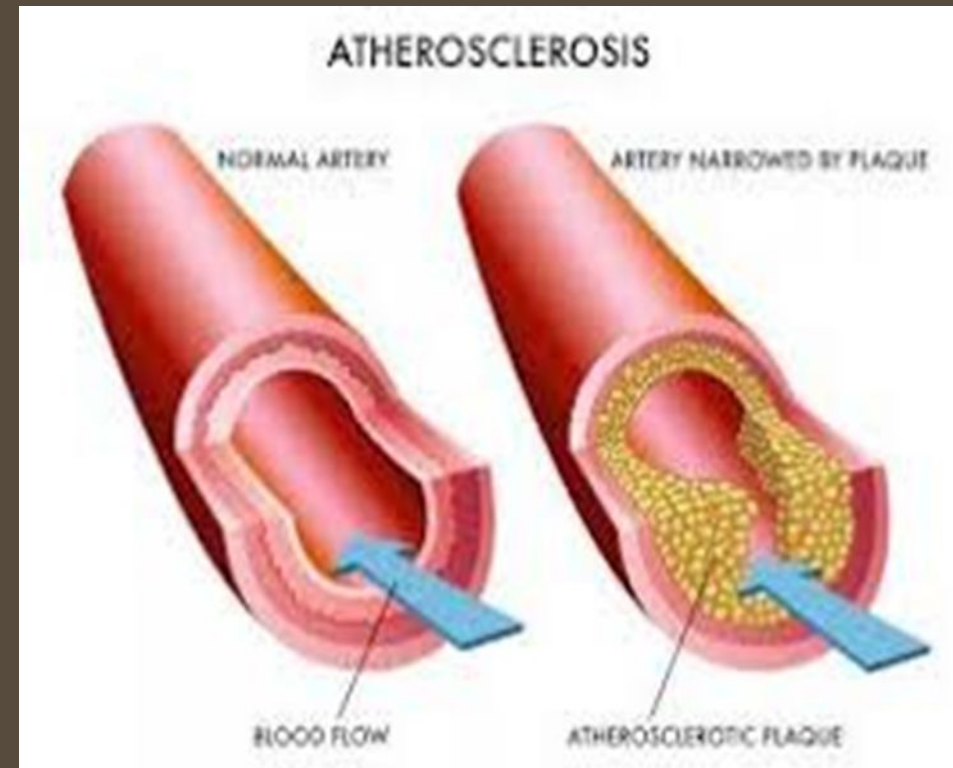
- Vitamin E (Tocopherol):** The "Front line" in the lipid membrane and "sacrifices" itself by giving an electron to a lipid radical, stopping the chain reaction.
- Vitamin C (Ascorbate):** The "Reload" agent stays in the watery part of the cell gives electrons back to Vitamin E so it can fight again. .

3.Repair (Enzymes):

- Phospholipase A2:** Cleaves out the damaged fatty acid so it can be replaced.

Lipoproteins include:

- a) Chylomicrons
- b) VLDL
- c) LDL
- d) HDL



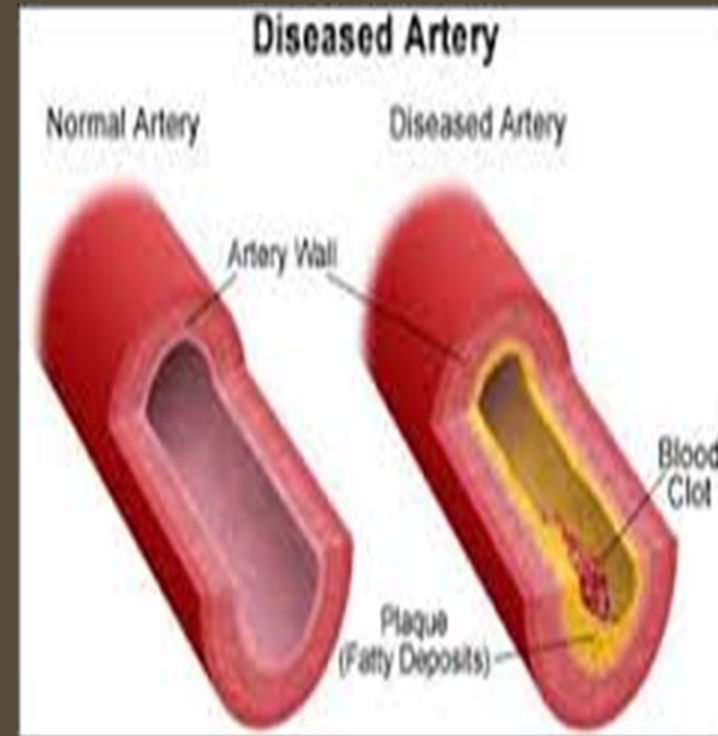
Coronary risk factors

- high cholesterol
- smoking
- older age
- family history
- male gender
- diabetes
- high blood pressure



RISKS ASSOCIATED WITH RAISED CHOLESTEROL

- Coronary artery disease (CAD)
- Cerebrovascular accident (CVA)
- Pancreatitis



RISK FACTORS FOR CAD

Low HDL Cholesterol (<40mg/dl)

Hypertension

Family history

Diabetes Mellitus

Cigarette smoking

Age (male 40 years older)

(female 55 years older)

CLINICAL IMPLICATION: ATHEROSCLEROSIS

- The LDL Modification:**

Native LDL is not particularly atherogenic. However, **Oxidized LDL (oxLDL)** is highly dangerous.

- Scavenger Receptors:**

Unlike normal LDL receptors, the scavenger receptors on macrophages (SR-A1) are not down-regulated.

- Foam Cells:**

Macrophages consume oxLDL until they become "Foam Cells," which accumulate to form the atherosclerotic plaque.

Increased cholesterol	Decreased cholesterol
Sedentary lifestyle	Increased exercise
Obesity	High fiber diet
Smoking	Diet low in simple sugars and fried food
Alcohol intake	Omega-3-fatty acids from fish, seed oils
Type 2 DM	
Genetic Factors	

CLINICAL IMPLICATION: OTHER DISORDERS

- **Neurodegeneration:** High lipid content in the brain (myelin) makes it susceptible to peroxidation, contributing to Alzheimer's and Parkinson's.
- **Liver Disease:** Peroxidation of hepatocyte membranes leads to enzyme leakage (ALT/AST) and contributes to Steatohepatitis.
- **Membrane Fluidity:** Peroxidation causes membranes to become rigid, affecting ion transport and cell signaling.

DEFENSE SYSTEMS (THE ANTIOXIDANT SHIELD)

- **Enzymatic:** * *Superoxide Dismutase (SOD)*: Neutralizes superoxide.
- *Glutathione Peroxidase*: Reduces lipid hydroperoxides to alcohols.
- **Non-Enzymatic:** * *Vitamin E*: Lipid-soluble chain breaker.
- *Vitamin C*: Water-soluble; regenerates Vitamin E.
- *Beta-carotene*: Scavenges singlet oxygen.

CONCLUSION

- Lipid peroxidation is a hallmark of oxidative stress.
- The transition from "Healthy Cholesterol" to "Oxidized LDL" is the pivotal step in cardiovascular disease.
- Maintenance of the antioxidant-oxidant balance is essential for cellular longevity.



Thank You!